GGR376: Spatial Data Science II

Examining Spatial Mismatches Through Automobility Traps

Janiri Nunal 1005003050
Gurman Sanga 1008195846
Tissan Kugathas 1006974026
Kashy Namboothiri 1007401740

April 6, 2026

Introduction 3

Research Question 4

Data Description & Study Area 4

Methods 6

Results 8

Discussion and Interpretation 13

Limitations and Uncertainty 15

Conclusion 16

References 17

Contribution Statement 20

Appendix 21

Introduction

In most urban regions, lower-income households are expected to rely more heavily on public transit and less on private vehicles to help ease financial constraints. In the Greater Toronto and Hamilton Area (GTHA), this expectation does not always hold as many significant parts of the region, rapid transit is simply too far away or too infrequent to be a realistic option which creates an opposite outcome where lower-income households may be structurally forced into car ownership not by preference, but by the geography of where they live relative to where transit exists (Mattioli et al., 2020; Lucas et al., 2016).

This is characterized as an automobility trap, which occurs when housing, employment, and regional rapid transit are spatially mismatched (Mattioli et al., 2020). When affordable transit alternatives are unavailable or unreasonable, car ownership shifts from optional to an obligation regardless of a household's financial capacity to afford it which carries serious financial consequences for lower-income households, as vehicle financing, insurance, fuel, and maintenance costs can consume an excessive amount of household income that could otherwise be spent on meaningful living costs (Mattioli, 2017; Lucas et al., 2016). The key point is that the automobility trap is not entirely an income problem, but a combination of income and a household's location relative to regional rapid transit infrastructure (Barri et al., 2021).

The GTHA makes this an important and well suited case to study as rapid suburban growth across Peel, York, Durham, and Hamilton regions, combined with rising housing costs in the City of Toronto, has progressively displaced lower-income households away from Toronto's transit-rich urban core toward peripheral municipalities where Metrolinx rapid transit coverage is limited, and employment centres are more dispersed (Hulchanski, 2010; Walks, 2012). This produces a core-periphery structure in which there is dense, rapid transit in the urban centre and sparse coverage at the suburban fringe, meaning that where a household is located within the region may be just as important as its income level in determining whether car ownership is a choice or a necessity thus may generate automobility-trap-like patterns in specific parts of the region, particularly where low household income and weak proximity to rapid transit coincide.

To analyze automobility traps in the GTHA, this study looks for patterns across 1,512 census tracts covering Toronto, Hamilton, and Oshawa, where car dependency is defined as the tract-level share of employed commuters who drive as their primary mode, and to link financial capacity to transportation behaviour, median household income is used as a key indicator, using data from the 2021 Canadian Census (Statistics Canada, 2021a). Transit accessibility is measured as the straight-line distance between each tract's geometric

centroid and the nearest station in the Metrolinx's Frequent Rapid Transit Network (FRTN), which includes GO Transit rail, Bus Rapid Transit, and Light Rail Transit lines. Focusing on the Metrolinx infrastructure serves as a primary regional-scale alternative to car commuting across the broader GTHA that local municipal networks such as the TTC cannot match, as they do not extend with comparable frequency or coverage into the suburban municipalities where car dependence is highest (Metrolinx, 2018). Using spatial analysis and machine learning to determine whether income and proximity to FRTN are associated with spatial patterns of car dependency in the GTHA, this project seeks to identify areas where structural conditions may create localized automobility traps.

Research Question:

To what extent do household income and proximity to Metrolinx's Frequent Rapid Transit Network explain census tract-level car dependency in the GTHA?

Data Description & Study Area

The major focus of the study lies in the cities of Toronto, Hamilton, and Oshawa Census Metropolitan Areas(CMAs) which make up the Greater Toronto and Hamilton (GTHA). The region serves as a critical landscape for understanding the automobility trap through the following datasets:

Socioeconomic and Behavioural Data
To assess financial capacity and commuting patterns, The 2021 Census of Canada from CHASS is used to obtain the variables. The median household income serves as the independent variable representing the financial capacity of households while the dependent variable is defined as the tract-level percentage of car commuters whose main mode is car, truck, or van as driver representing regional mobility patterns (Statistics Canada, 2021a).

Spatial boundaries and weights
The spatial analysis is conducted at the Census Tract (CT) level by using the 2021 cartographic boundary files(Statistics Canada, 2021b). The final dataset contains 1,512 Census Tracts and all geometries projected to UTM Zone 17 (EPSG: 32617) ensuring accurate Euclidean distance calculations.

Infrastructure Data
The dataset is limited to incorporating provincial Metrolinx datasets, specifically the Frequent Rapid Transit Network (FRTN) that intentionally excludes standard municipal bus routes to focus on infrastructure that provides an alternative to driving(Metrolinx, 2022). The FRTN data is defined by 382 stations and 124 line features representing the regional rapid-transit network. A proximity variable Dist_KM was constructed by calculating the Euclidean distance

Software, Libraries and Computational Environment

All spatial and data operations are performed in Python using a Google Colab environment. Geopandas is used for data manipulation and coordinate transformation while mapclassify is useful for map classification. The construction of weights and Exploratory Spatial Data Analysis (ESDA) was done using libpysal and esda. Machine learning and statistical modeling was performed with statsmodels for regression, XGBoost for predictive modeling, and Geoshapely for interpreting the spatial feature importance.

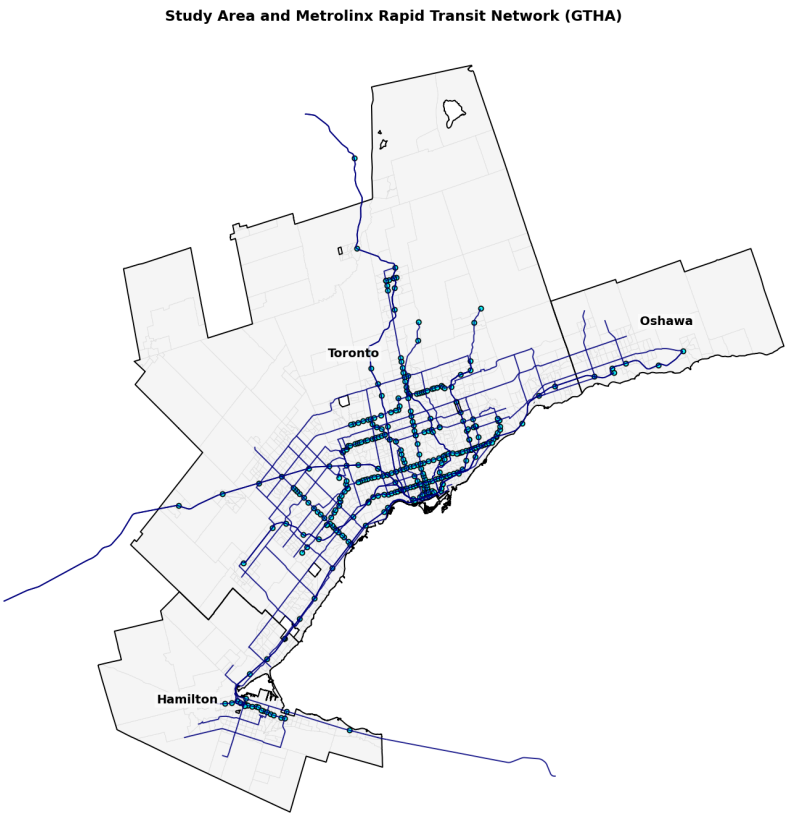


Figure 1. Study area and Metrolinx rapid-transit network across the Toronto, Hamilton, and Oshawa Census Metropolitan Areas. Census tracts define the analytical study region, while the network layer shows the broader rapid-transit system used to calculate tract-level proximity.

Methods

Data Preparation and Variable Construction

Throughout the project, all data processing, spatial analysis, and modelling were performed in Python 3.12, and sanity checks were performed to verify that the data did not include impossible or suspicious values that could distort the results.

Census data were obtained from CHASS (Statistics Canada, 2021a) and filtered to the Toronto, Hamilton, and Oshawa CMAs using GeoPandas (GeoPandas developers. 2024), and with the census data, tract-level car dependency (Pct_Car_Commute) was computed as the number of employed commuters who drive as their primary mode of transportation (Car_Drivers) divided by the total number of commuters (Total_Commuters), multiplied by 100. Also from the census, Median household income (Median_Income) served as the primary socioeconomic predictor.

Using Metrolinx's FRTN Station and Transit lines data (Metrolinx, 2022), the layers were reprojected to the same projection used in the census data, which was UTM Zone 17N (EPSG:32617), prior to performing any distance operations to ensure metric accuracy and using GeoPandas (GeoPandas developers, 2024), the straight-line Euclidean distance in kilometres between each tract's geometric centroid and the nearest Metrolinx FRTN station was computed and recorded as Dist_KM, which will be used as the transit accessibility predictor. The cleaned attributes were joined to 2021 Statistics Canada census tract polygons (Statistics Canada, 2021b) using the shared identifier CTUID, resulting in a final dataset containing 1,512 tracts.

Spatial Weights Matrix

A Queen contiguity spatial weights matrix was constructed from the 1,512 tract polygons using libpysal (Rey & Anselin, 2007) and row-standardized before use in any spatial statistic. One island tract with no Queen neighbours was identified and retained in the dataset, but contributed no spatial lag in any autocorrelation calculation.

Exploratory Spatial Data Analysis

Global spatial autocorrelation in Pct_Car_Commute was calculated using Global Moran's I with 999 conditional permutations via esda (Rey & Anselin, 2007) to determine whether car dependency is spatially clustered across the GTHA, then the local cluster structure was identified using Local Indicators of Spatial Association (LISA; Anselin, 1995) via esda. Raw p-values were corrected for multiple comparisons using the Benjamini–Hochberg false discovery rate (FDR) procedure (Benjamini & Hochberg, 1995), and only tracts surviving FDR correction are reported in the results.

Baseline OLS and Residual Diagnostics

Statsmodels (Seabold & Perktold, 2010) was used to perform an Ordinary Least Squares (OLS) regression with Pct_Car_Commute as the dependent variable and Median_Income and Dist_KM as predictors. Multicollinearity was assessed using variance inflation factors. To test whether OLS residuals exhibited remaining spatial structure, following the standard OLS-SOP diagnostic approach, Global Moran's I was applied to the model residuals under 999 permutations using esda (Rey & Anselin, 2007). A statistically significant residual Moran's I suggests that the linear model fails to account for spatial dependence in the data, implying the need for a more flexible modelling approach.

Spatially Aware XGBoost

Given the formal evidence of residual spatial autocorrelation in the OLS model as well as the expected nonlinear relationships between the predictors and car dependency, an XGBoost gradient-boosted tree model was estimated using the XGBoost library (Chen & Guestrin, 2016). XGBoost was chosen for its ability to capture nonlinear feature interactions and threshold effects that a linear model cannot handle, as well as its compatibility with the GeoShapley decomposition used in the next stage, and to avoid overfitting, the model was set up with 400 estimators, a maximum tree depth of 4, a learning rate of 0.05, and subsampling and L2 regularization. The features included Median_Income, Dist_KM, and the longitude and latitude of each tract centroid, allowing the model to learn residual location-specific patterns.

Model performance was evaluated under two schemes using scikit-learn (Pedregosa et al., 2011). A random 80/20 train–test split was used as a baseline, and a spatial holdout design was then applied, in which geographically contiguous blocks of tracts were reserved for testing to prevent information leakage from spatially adjacent training observations (Pohjankukka et al., 2017). The spatial holdout result is treated as the primary honest performance benchmark, and the random-split result is reported for reference only and should not be interpreted as the model's true generalization ability.

GeoShapley Decomposition

Following spatial validation, the XGBoost model was refitted on the full dataset to maximize available information for interpretation and GeoShapley values were then computed using the GeoShapley library (Li, 2024) to decompose each tract's fitted prediction into four additive components, which are an intrinsic geographic effect (GEO), a non-spatial income effect, a location-modulated income interaction (Median_Income × GEO), and a location-modulated transit-distance interaction (Dist_KM × GEO). GeoShapley contributions are expressed in the same units as the outcome variable and should be interpreted as marginal contributions to the model prediction, not as causal effects. The decomposition was verified by confirming that the four components reconstruct each tract's fitted value to within a negligible margin of error.

Heuristic Synthesis of Candidate Automobility-Trap Zones

The final step was creating a heuristic synthesis map using GeoPandas (GeoPandas developers, 2024) to identify potential candidate automobility-trap zones where a tract has to meet two gate conditions at once

(below-median household income and above-median distance to the nearest FRTN station), which apply the automobility trap concept's basic structural preconditions (financial constraints combined with a lack of proximity to regional rapid transit). The support score for qualifying was computed using four binary indicators (high observed car dependency, high model-predicted car dependency, positive intrinsic GEO effect, and positive GEO × transit distance interaction), where the tracts were placed into one of three categories based on their support score (Weak Candidate, Candidate, and Strong Candidate), with higher tiers suggesting more evidence across multiple indicators. Using multiple binary indicators to categorize candidates avoids overclassification from any single measure. A key point to remember is that identified zones should be treated as areas that may need further policy attention than confirmed cases of household-level automobility traps (Mattioli, 2017).

Results

Descriptive and ESDA results

Across the 1,512 census tracts in the study area, distance from each tract centroid to the nearest Metrolinx station ranges from 0.05 km to 42.19 km, with a median of 1.59 km and a mean of 2.78 km. These numbers alone show that some tracts are much closer to rapid transit than others.

The ESDA maps suggest a strong centre-periphery contrast, where lower car dependency is concentrated in Toronto's core. Higher car dependency is more common across much of the suburban and outer region. The distance map follows a similar pattern. Tracts that sit farther from the rapid-transit backbone are more common outside the core. The income is uneven too, but it does not match the other two maps in a clean way, so the pattern does not reduce to one simple divide.

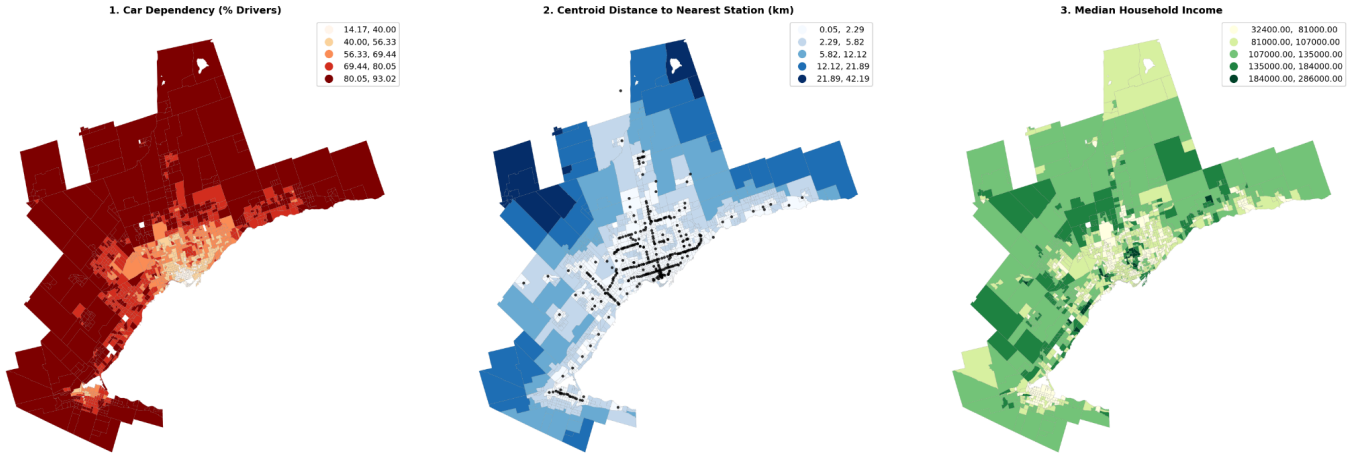


Figure 2. Exploratory spatial distribution of tract-level car dependency, rapid-transit proximity, and median household income across the GTHA.

Global and local spatial structure

The spatial clustering is strong. Global Moran's I for tract-level car dependency is 0.8815, with an expected value of -0.0007, a permutation p-value of 0.0010, and a z-score of 60.3136. This is an extremely high positive result. Tracts with similar levels of car dependency are located near each other far more than spatial randomness would suggest. This is also clear in Appendix Figure A1, where the observed Moran's I lies far beyond the permuted null distribution.

The local results show the same pattern at a finer scale. After false discovery rate correction, 676 tracts remain significant. Of these, 384 are High-High, 289 are Low-Low, and 3 are High-Low, while 836 tracts are not significant. These are not just isolated tracts. They form larger zones. High-High tracts dominate much of the outer region, while Low-Low tracts remain concentrated in the Toronto core. There are no Low-High tracts in the final corrected result.
A High-High tract is a tract with relatively high car dependency surrounded by neighbouring tracts with similarly high values. It is not, by itself, proof of an automobility trap. At this stage, it is better read as a statistically significant concentration of car dependency.

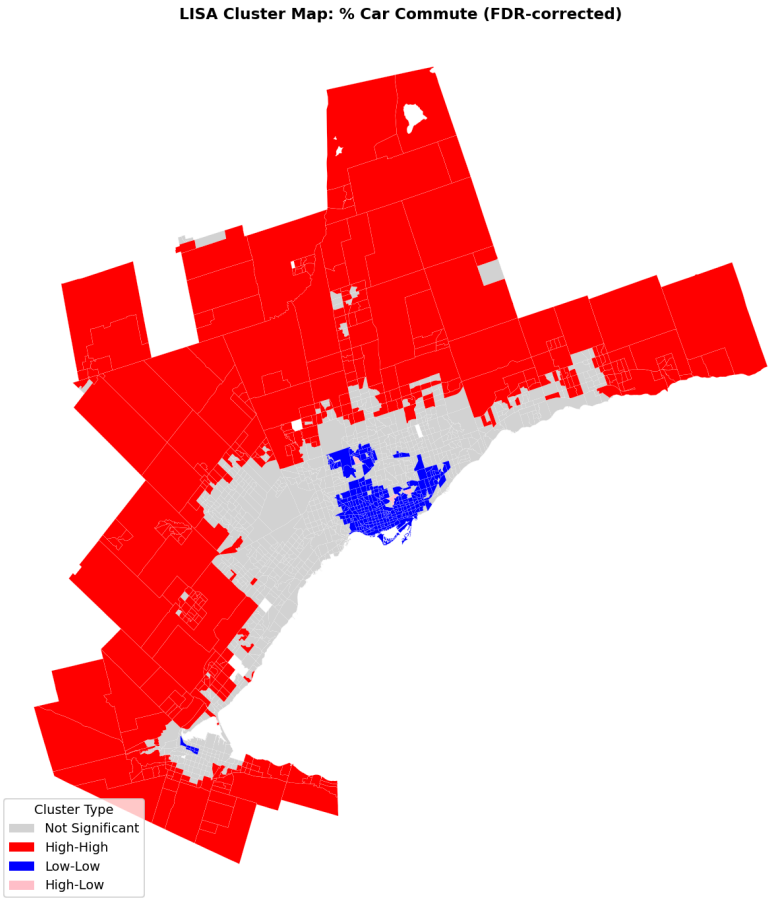


Figure 3. FDR-corrected LISA cluster map of tract-level car dependency in the GTHA. High-High and Low-Low classes identify statistically meaningful local concentrations rather than simple visual patterning.

Baseline OLS and residual structure

The baseline OLS model uses Median_Income and Dist_KM to predict tract-level car dependency. Multicollinearity is low and both predictors are statistically significant, with variance inflation factors of 1.066 for both variables. The model fit is $R^2 = 0.3355$, so it picks up part of the pattern, but a lot is still left over.

The coefficient for Dist_KM is 1.2281. For tracts at the same income level, being 1 km farther from the nearest Metrolinx station is associated with about 1.23 percentage points more car commuting. The coefficient for Median_Income is 0.0002. Put more simply, that is about 2 additional percentage points of car commuting per $10,000 increase in tract median income, for tracts with distance as a constant. So the pattern is more complicated than just saying lower income means more driving. Many higher-income suburban tracts are also highly car-dependent.

What matters more here is the residual pattern. Moran's I on the OLS residuals is 0.8394, with a permutation p-value of 0.0010. The residuals remain strongly spatially clustered. The residual map makes this visible. The model is still leaving major geographic structure behind. That is the main limit of the OLS stage. Appendix Figure A2 reinforces this point by showing that the observed residual Moran's I also lies far beyond the permuted null distribution.

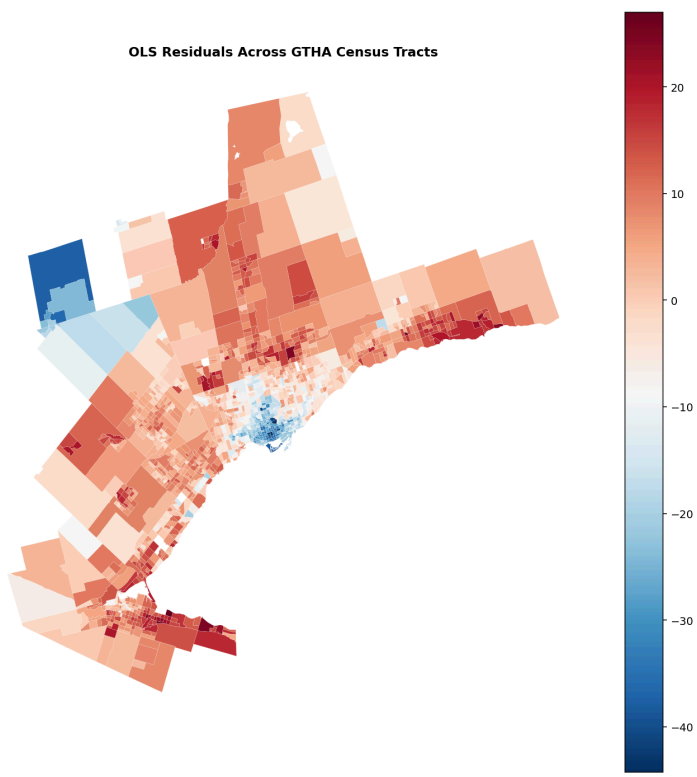


Figure 4. OLS residual map showing the spatial structure that remains after controlling for median household income and rapid-transit distance.

Spatially aware XGBoost

XGBoost does much better than OLS, but depending on how the model is tested the results change. Under a random split test, the model reached $R^2 = 0.9104$, with RMSE = 4.8606, and MAE = 3.5871. These values are strong, but they are also optimistic because nearby tracts can appear in both training and test sets.

The spatial split is the score that matters more here. When separate geographic blocks are held out, the model reaches an R^2 value of 0.7769, RMSE of 6.1527, and an MAE value, 4.8053. Even with that stricter test, it still does much better than the OLS baseline. The drop from the random split to the spatial split is expected. It shows that ordinary validation was overstating performance. Appendix Figure A3 shows the spatial holdout design itself, making clear that the training and test areas were geographically distinct. Table 1 shows the difference between the OLS baseline and both XGBoost results. The nonlinear model is picking up much more of the pattern in tract-level car dependency. That is the main reason the spatially aware XGBoost model is more useful here than the baseline linear model. A supplementary observed-versus-predicted fit diagnostic is also shown in Appendix Figure A4.

Model	**Evaluation**	**R^2**	**RMSE**	**MAE**
OLS baseline	Full sample	0.3355	13.0069	10.0434
XGBoost	Random test split	0.9104	4.8606	3.5871
XGBoost	Spatial holdout test	0.7769	6.1527	4.8053

Table 1. Comparison of baseline OLS and XGBoost model performance under random and spatially aware validation.

Note. OLS is reported as a full-sample explanatory benchmark, while XGBoost performance is reported on held-out test sets. The spatial holdout score is the main predictive benchmark because it reduces leakage from nearby tracts.

GeoShapley interpretation

GeoShapley was used to interpret the final XGBoost model. The base value is 69.1303, meaning the model begins from an average predicted car dependency of about 69.1% before tract-specific effects are added. The reconstruction check is near exact, with a maximum absolute difference of about 1.53×10^{-5}.

The strongest contribution in the final model is the intrinsic location effect, GEO. Its range is wider than the range of the non-spatial main effects. So location itself is carrying a lot of the prediction. A tract's place in the broader regional structure can push predicted car dependency up or down even after income and rapid-transit distance are already included. This is also summarized in Appendix Figure A5, where GEO ranks above the non-spatial feature contributions.

The two interaction maps do not behave the same way. The GEO × Dist_KM map is easier to read as a regional pattern than the GEO × Income map. The map suggests that geography modifies the effect of rapid-transit distance in a more regionally consistent way than it modifies the effect of income.

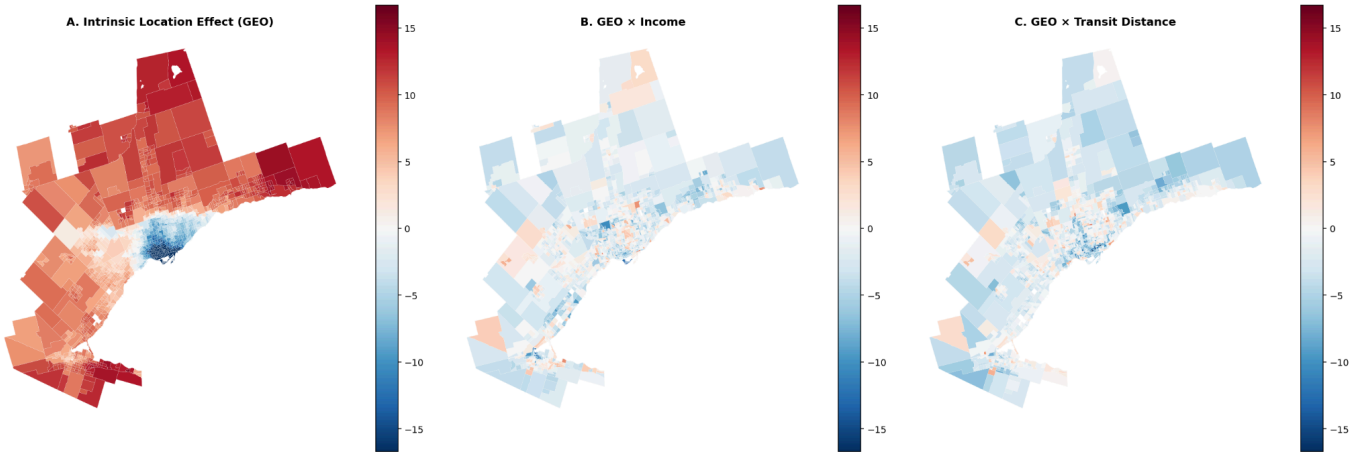


Figure 5. GeoShapley decomposition of the final XGBoost model, showing the intrinsic location effect and location-specific interactions with income and rapid-transit distance across the GTHA.

Final heuristic synthesis

The final synthesis is intentionally strict. A tract first had to meet two gate conditions. It had to be relatively lower income and relatively poorly served by rapid transit. Among these tracts, additional support came from high observed car dependency, high predicted car dependency, positive intrinsic location effect, and positive GEO × Dist_KM contribution.

Using that rule, the final counts are 1453 Not Candidate, 30 Weak Candidate, 4 Candidate, and 25 Strong Candidate. The classification is selective. Most tracts are not identified as candidate zones.

The strong candidates are unevenly spread across the study area, with 7 in Hamilton, 6 in Oshawa, and 12 in Toronto. The remaining candidate tracts are split between Hamilton and Toronto. Weak candidates are more dispersed, with 10 in Hamilton, 7 in Oshawa, and 13 in Toronto. This pattern suggests that only a relatively small set of tracts satisfy the stricter combination of lower income, weak rapid-transit proximity, high car dependency, and reinforcing spatial effects. The map should still be read as a heuristic synthesis, not as proof of household-level coercion.

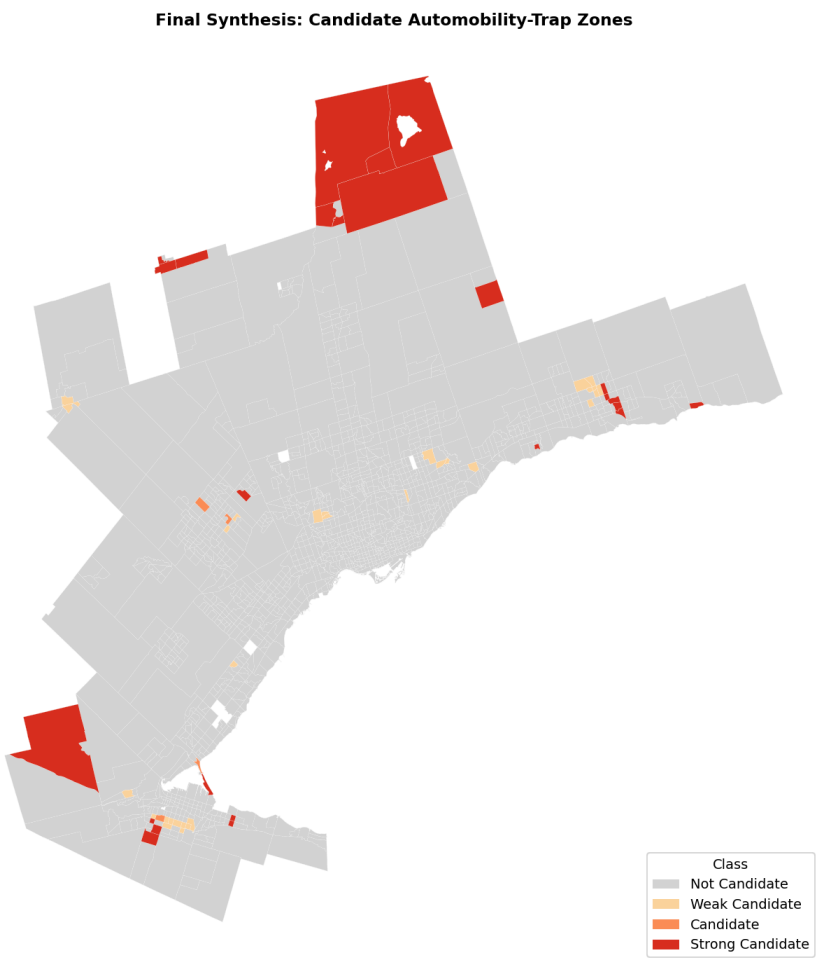


Figure 6. Final heuristic synthesis map identifying candidate automobility-trap zones under the study's strict classification rule.

Discussion and Interpretation

The main result of this project is that car dependency in the GTHA is not just uneven. It is spatially organized in a very strong way. The ESDA maps suggested that early on, but the autocorrelation results made it much harder to treat as just a visual impression. Lower car dependency stays concentrated in the Toronto core, while higher car dependency dominates much of the outer region. So geography is not just the setting here. It is part of what is producing the pattern. The strength of that clustering is also visible in Appendix Figure A1, where the observed Global Moran's I sits far beyond the permuted null distribution.

The OLS stage helps show this clearly. Income and rapid-transit distance both matter, and the model does pick up part of the variation. But that is not the main takeaway from that section. The more important point is that the residuals remain heavily clustered. A simple linear model using only income and distance still leaves major geographic structure behind. That does not mean OLS was useless. It means the spatial pattern is

stronger than what a basic two-variable model can absorb. Appendix Figure A2 supports that point directly, showing that the observed residual Moran's I also lies far beyond the permuted null distribution.
That is also why the move to XGBoost matters. The point is not just that the numbers get better. The point is that once the model is allowed to pick up nonlinear structure, a lot more of the tract-level pattern becomes explainable. Even under the stricter spatial holdout, the model still performs much better than OLS. The random split score looks stronger, but that is exactly why the spatial split is the more useful one here. It forces the model to predict into different parts of the region rather than into nearby tracts that already look similar. So the drop from the random split to the spatial split is not a weakness in the project. It actually confirms that the validation was handled more honestly. The spatial holdout design itself is shown in the Appendix Figure A3, and the fit of the final XGBoost model can be seen more directly in Appendix Figure A4.

Income also needs to be read carefully. The results do not support a simple story where lower income always means more driving. In fact, many higher-income suburban tracts are highly car-dependent too. That matters because it changes how income should be interpreted. In this project, income works better as a marker of vulnerability than as a full explanation on its own. High car dependency in a high-income tract does not mean the same thing as a high car dependency in a lower-income tract.

The GeoShapley results push the interpretation further. The strongest contribution in the final model is not income by itself and not distance by itself, but location. That is a strong finding. It means where a tract sits in the wider regional structure still matters a great deal even after the main non-spatial variables are already included. The distance interaction appears to matter more than the income interaction too. The GEO × Dist_KM map is easier to read as a regional pattern, while the GEO × Income map is more patchy. So the clearest mechanism in the final model is not just weak rapid-transit proximity on its own. It is weak rapid-transit proximity being reinforced in some places more than others by the geography around it. That broader contribution pattern is also summarised in Appendix Figure A5. The nonlinear feature relationships are shown more directly in Appendix Figure A6.

That matters for the final synthesis map. The map is strict, and that is one of its strengths. Most tracts are not identified as candidate zones. Only a relatively small number satisfy the full combination of lower income, poor rapid-transit proximity, high car dependency, and reinforcing spatial effects. That selectiveness actually makes the final map more believable. If almost the whole outer region had been labeled as a candidate zone, the result would have been much less convincing.

At the same time, the project still has to stay careful about what it is actually claiming. The final map is a heuristic synthesis, not direct proof of household-level coercion. The model works at the tract level, and the variables are also measured at the tract level. What the project can show is that some tracts appear more exposed than others to a structural combination of weak regional rapid-transit proximity and high car dependency, especially where that pattern overlaps with lower income. What it cannot show is that every household in those tracts experiences transport pressure in the same way.

That is also where the biggest real-world limit shows up. Distance to the nearest station is still a proxy. It is a useful one, but it is still just a straight-line centroid-to-station distance. It does not include local transit frequency, transfer burden, walking conditions, or service timing. So a tract can appear reasonably close to the regional network in geometric terms and still be poorly connected in practice. That does not undo the current findings, but it does shape how far the interpretation should go.

So the broad story here is not just that distance matters, or that income matters, or that a more flexible model performed better. The stronger conclusion is that tract-level car dependency in the GTHA is structured by geography in a way that a simple linear model cannot capture very well. Once space is handled more seriously, the clearest signal is the combination of location and the way location modifies the penalty of weak rapid-transit proximity. That is the part of the project that most strongly supports the automobility trap framing.

Limitations and Uncertainty

Although our results yielded good results with strong spatial patterns identified, there are several limitations our study must acknowledge. First, the analysis is conducted at the Census Tract level which is therefore subject to the ecological fallacy, meaning tract-level results are not enough to depict individual household behaviour and cannot be interpreted as direct evidence. Second, our results are also affected by the Modifiable Areal Unit Problem (MAUP), as spatial clustering patterns with a Global Moran's I value of 0.8815 however this value may vary depending on the scale or configuration of tract boundaries.

Our third limitation is that transit accessibility is measured using Euclidean distance (Dist_KM) from tract centroids to the nearest rapid transit station. This is a limitation due to simplifying paths travelled in order to be computationally efficient. This approach takes real world travel conditions and dismisses factors like road network routing, travel time, or physical barriers. Due to the focus of the study we only use regional rapid transit and exclude the TTC services such as bus routes and service frequency as we are trying to identify automobile traps.

The fourth limitation is that spatial clustering results are dependent on the specification of Queen contiguity weights. This means that the Global Moran's I and LISA outputs are sensitive to how neighborhood relationships are defined. Our model evaluation is also limited as we split our data by geographical location which holds us accountable to a single spatial holdout validation. This model yields better results than repeated spatial cross-validation, which may influence performance estimates but is a more robust model.

Lastly the GeoShapley interpretation is only an explanation rather than causal as it does not isolate for cause and effect, account for hidden variables and does not prove real world mechanisms. It ultimately only identifies model contributions without proving causal mechanisms. As we used a strict heuristic classification to identify automobility trap zones to classify our areas which represents a structured analytical inference it is

best to understand that it is not a definitive classification. These limitations do not invalidate the results, but they define the scale and strength of the conclusions that can reasonably be drawn.

Conclusion

This study concludes that tract-level car dependency in the Greater Toronto and Hamilton Area is strongly spatially clustered. The evidence exhibits a very high Global Moran's I value at 0.8815 with a clear High-High and Low-Low clustering pattern. Car dependency is concentrated mainly across the suburban and peripheral regions, while lower dependency remains tightly clustered within the downtown core. This clearly exhibits that mobility patterns are shaped by underlying spatial structure and is not due to randomness.

Our results demonstrate that household income and proximity to Metrolinx's Frequent Rapid Transit Network meaningfully explain census tract-level car dependency across the GTHA. With our model evaluation the baseline OLS model captures part of this relationship with the spatially aware XGBoost model revealing a much stronger explanatory power. This indicates that car dependency is shaped by complex, non-linear and spatially structured processes. Put together, our models confirm that both income and transit accessibility are significant determinants of car dependency at the neighbourhood scale.

The findings further indicate that the biggest factor is the proximity to rapid transit, the further they are, the higher car dependency is. Location is consistently reinforcing its effect across the region. Income contributes to this relationship but operates more unevenly across space. Overall, the results provide strong evidence that spatial configuration of transit networks and socioeconomic conditions jointly structure car dependency, directly addressing the research question and confirming the presence of localized mobility constraints within the GTHA.

References

Anselin, L. (1995). Local indicators of Spatial Association—lisa. *Geographical Analysis*, *27*(2), 93–115. https://doi.org/10.1111/j.1538-4632.1995.tb00338.x

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Chen, T., & Guestrin, C. (2016). XGBoost. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785

GeoPandas developers. (2024). GeoPandasVersion (1.1.3). *GeoPandas Documentation*. Retrieved from https://geopandas.org/en/stable/docs.html.

Hulchanski, D. J. (2010). *The three cities within Toronto: Income Polarization among Toronto's neighbourhoods, 1970-2005*. The Three Cities Within Toronto: Income Polarization Among Toronto's Neighbourhoods, 1970-2005. https://utoronto.scholaris.ca/items/b7a2e840-0e5f-4be3-a307-70662fbafd9b

Li, Z. (2024). Geoshapley: A game theory approach to measuring spatial effects in machine learning models. *Annals of the American Association of Geographers*, *114*(7), 1365–1385. https://doi.org/10.1080/24694452.2024.2350982

Lucas, K., Mattioli, G., Verlinghieri, E., & Guzman, A. (2016). Transport poverty and its adverse social consequences. *Proceedings of the Institution of Civil Engineers - Transport*, *169*(6), 353–365. https://doi.org/10.1680/jtran.15.00073

Mattioli, G. (2017). ‘forced car ownership’ in the UK and Germany: Socio-spatial patterns and potential economic stress impacts. *Social Inclusion*, *5*(4), 147–160. https://doi.org/10.17645/si.v5i4.1081

Mattioli, G., Roberts, C., Steinberger, J. K., & Brown, A. (2020). The political economy of car dependence: A systems of provision approach. *Energy Research & Social Science*, *66*, 101486. https://doi.org/10.1016/j.erss.2020.101486

Metrolinx. (2018). *Transit Access and Social Equity in the Greater Toronto and Hamilton Area*. Retrieved from https://assets.metrolinx.com/image/upload/v1663240133/Documents/Metrolinx/Transit_Access_and_Equity_2018.pdf.

Metrolinx. (2022). *Frequent rapid transit network [Geospatial dataset]*. https://www.metrolinx.com/en/about-us/open-data

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011, January 1). *Scikit-Learn: Machine learning in Python*. Journal of Machine Learning Research. https://jmlr.org/papers/v12/pedregosa11a.html

Pohjankukka, J., Pahikkala, T., Nevalainen, P., & Heikkonen, J. (2017). Estimating the prediction performance of spatial models via spatial K-fold cross validation. *International Journal of Geographical Information Science*, *31*(10), 2001–2019. https://doi.org/10.1080/13658816.2017.1346255

Rey, S. J., & Anselin, L. (2007). PySAL: A python library of spatial analytical methods. *Review of Regional Studies*, *37*(1). https://doi.org/10.52324/001c.8285

Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with python. *Proceedings of the Python in Science Conference*, 92–96. https://doi.org/10.25080/majora-92bf1922-011

Statistics Canada. (2021a). *2021 Census of Canada [Dataset]*. Chass Data Centre. https://datacentre.chass.utoronto.ca/

Statistics Canada. (2021b). *2021 census cartographic boundary files – Census tracts [Dataset]*. Statscan. https://www12.statcan.gc.ca/census-recensement/2021/geo/sip-pis/boundary-limites/index2021-eng.cfm?year=21

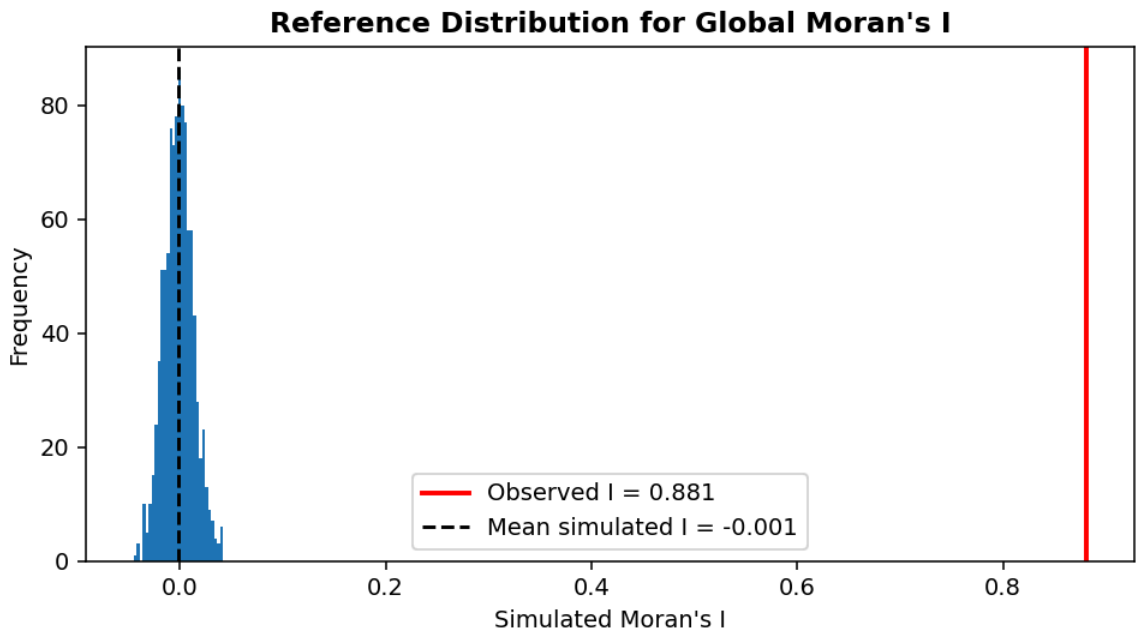
Walks, A. (2012). Suburbanism as a way of life, slight return. *Urban Studies*, *50*(8), 1471–1488. https://doi.org/10.1177/0042098012462610

Yousefzadeh Barri, E., Farber, S., Kramer, A., Jahanshahi, H., Allen, J., & Beyazit, E. (2021). Can transit investments in low-income neighbourhoods increase transit use? exploring the nexus of income, car-ownership, and Transit Accessibility in Toronto. *Transportation Research Part D: Transport and Environment*, *95*, 102849. https://doi.org/10.1016/j.trd.2021.102849

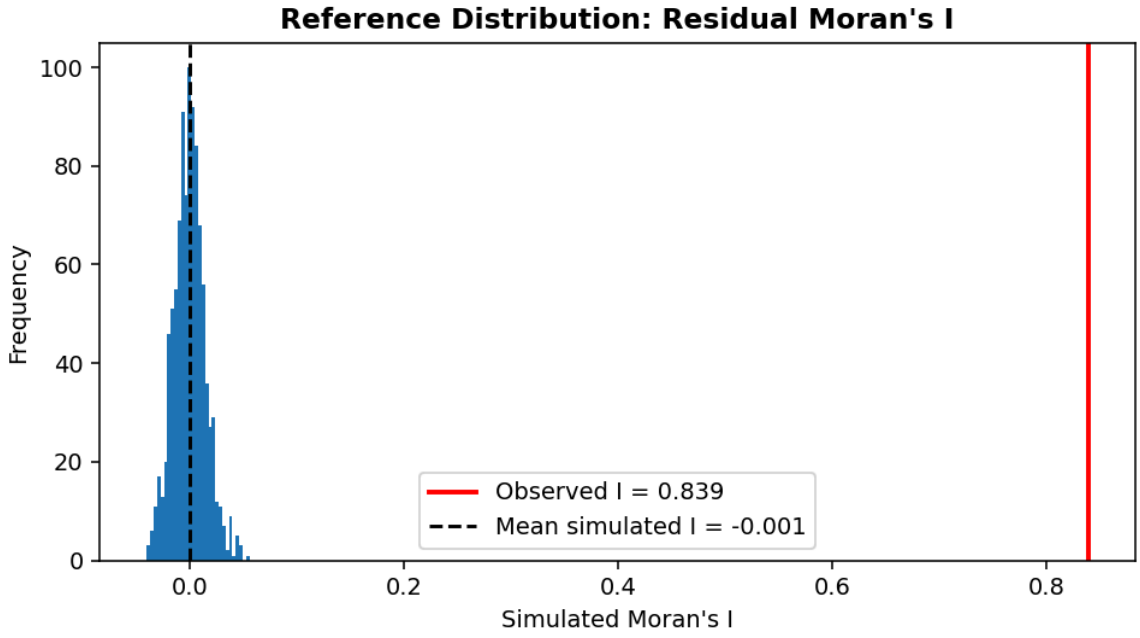
Contribution Statement

Group Member	Primary Work	Secondary Work
Janiri Nunal	● Worked on limitations and conclusions throughout the project ● Constructed the presentation slides	● Refined each member's part ● Took the data from the notebook and imputed it on the slides
Gurman Sanga	● Worked on Data & Study throughout the entire project ● Compiled Data for the notebook	● Refined each member's part ● Imputed the data into the notebook
Tissan Kugathas	● Worked on the Introduction and methodology throughout the project ● Constructed the code in the notebook	● Refined each member's part ● In partnership with Kashy, worked on analyzing data in the notebook
Kashy Namboothiri	● Worked on the Results and interpretation throughout the entire project ● Constructed the code in the notebook	● Refined each member's part ● In partnership with Tissan, worked on analyzing data on the notebook. Cleaned up the notebook.

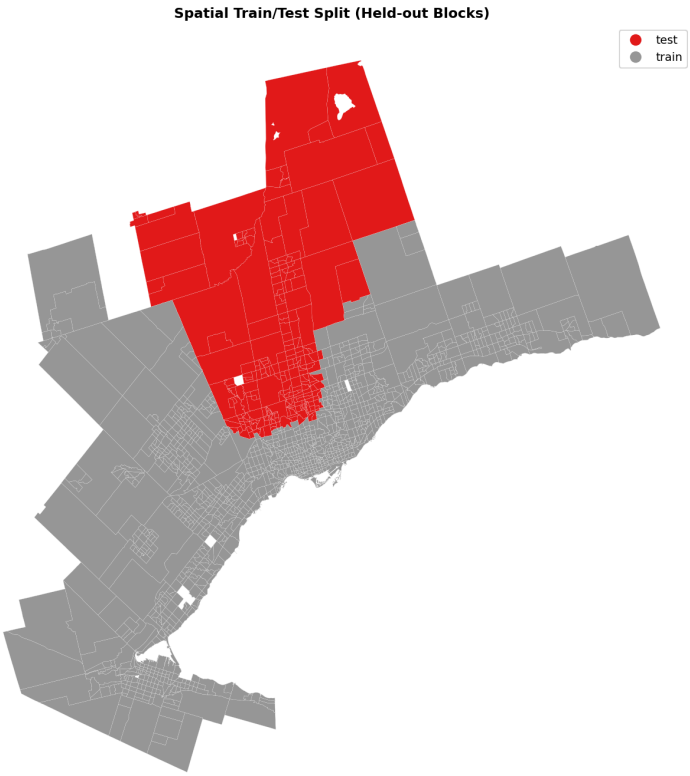
Appendix



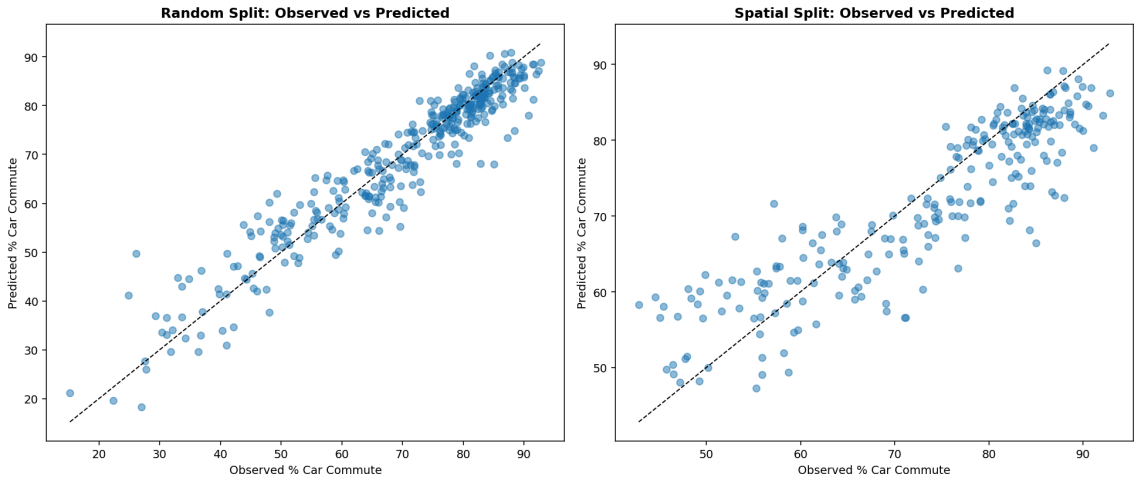
Appendix Figure A1. Reference distribution for Global Moran's I, showing that the observed spatial autocorrelation in tract-level car dependency lies far beyond the permuted null distribution.



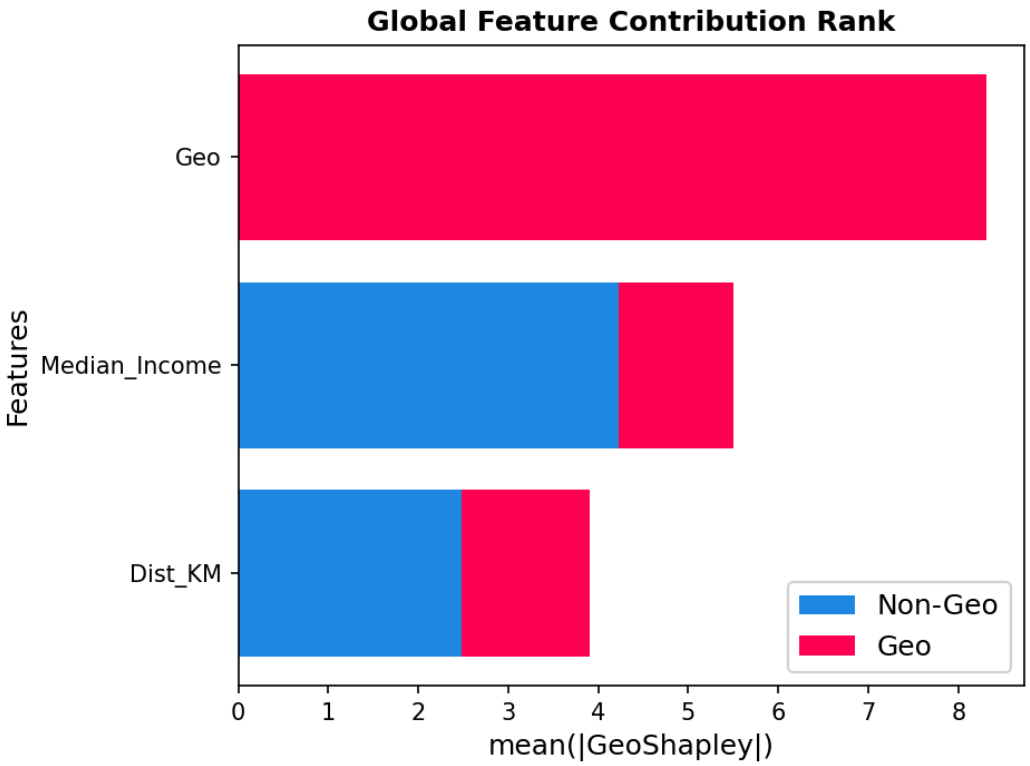
Appendix Figure A2. Reference distribution for residual Moran's I after the baseline OLS model, showing that substantial spatial autocorrelation remains in the model errors.



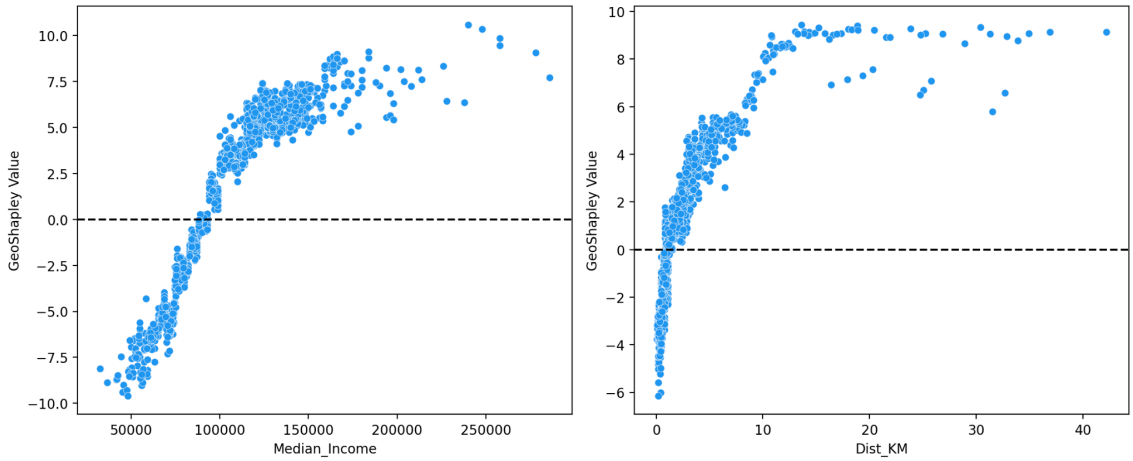
Appendix Figure A3. Spatial holdout design used for model validation, showing the geographically distinct blocks used to separate training and test areas.



Appendix Figure A4. Observed versus predicted values for the XGBoost model, included as a supplementary diagnostic of predictive fit.



Appendix Figure A5. Global contribution ranking from the GeoShapley decomposition, showing the relative importance of intrinsic location and non-spatial features in the final model.



Appendix Figure A6. Partial dependence plots for the primary feature effects, showing the nonlinear relationship between the main predictors and modelled car dependency.